CASE STUDY: "AR KITCHEN" 3D CULINARY SIMULATION

Role:	Mobile AR Developer & Pedagogical Designer
Tech Stack:	Unity 3D, Vuforia Engine (AR), C#, UI/UX Design
Target Users:	10th Grade Vocational School Students (SMK Tata Boga)
Subject:	Dasar Masakan Kontinental (Continental Culinary Basics)

1. Project Overview

Vocational High School (SMK) students majoring in Culinary Arts (Tata Boga) require extensive visual and practical references to master complex plating techniques and international dish standards. To solve the issue of high ingredient costs for constant physical practice, I developed **AR Menu**—a comprehensive "Modul Ajar" (Teaching Module) powered by Augmented Reality. This ecosystem transforms a traditional physical textbook into an interactive mobile experience, complete with 3D dish projections, digital learning materials, and an integrated quiz system.

2. Key Engineering & Design Challenges

A. Immersive Visualizations on a Budget

The Challenge: Culinary students need to inspect dish components, textures, and plating arrangements from all angles. Physical textbooks only provide flat 2D images, which are insufficient for grasping the spatial layout of continental dishes.

The Solution: Using the Vuforia Engine within Unity 3D, I programmed an AR scanner that tracks specific marker pages within the physical module. Upon detection, the app renders a highly detailed, 360-degree 3D model of the dish directly onto the book. This allows students to physically move their phones around the "virtual plate" to study the garnishing and structure without requiring expensive real-world ingredients.

B. Building a Complete Pedagogical Ecosystem

The Challenge: An AR gimmick is not enough for an educational setting; the application needed to function as a standalone, curriculum-aligned teaching tool that could assess student comprehension.

The Solution: Rather than just a camera app, I developed a full application architecture with three core modules: **Materi** (digital reading materials complementing the physical book), **Start** (the AR tracking camera), and **Evaluasi** (an interactive, gamified quiz system). Using C# scripting, I built a state-driven

multiple-choice evaluation system that provides immediate feedback, turning passive reading into active learning.

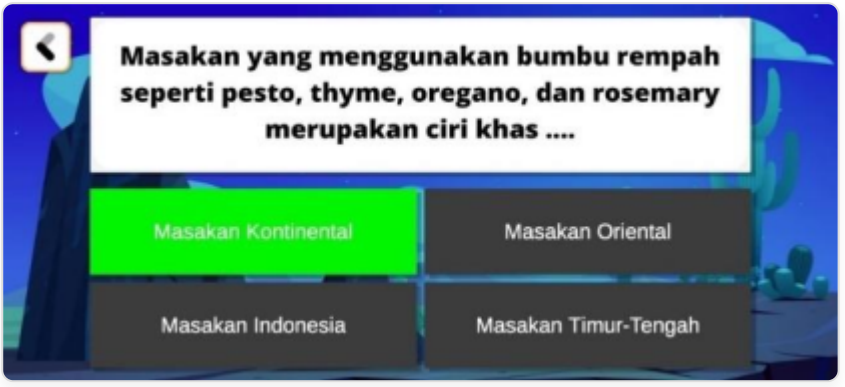
3. Application Interface & Output



Augmented Reality 3D Plating Simulation: Scanning the physical module to project a 360-degree interactive view of Continental dishes.



The Mobile Ecosystem: A complete pedagogical tool featuring Digital Modules, an AR Scanner, and Interactive Quizzes.



Interactive Gamified Evaluation: Real-time assessment system programmed in C# to validate student comprehension.